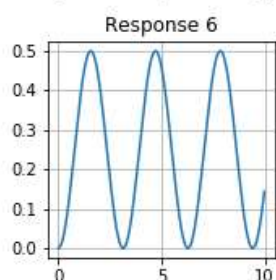
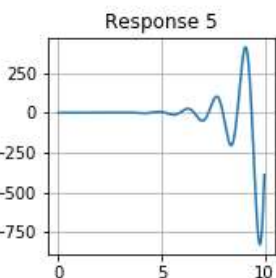
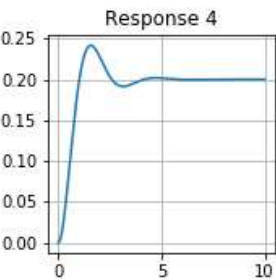
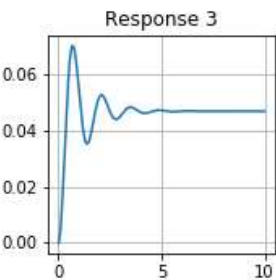
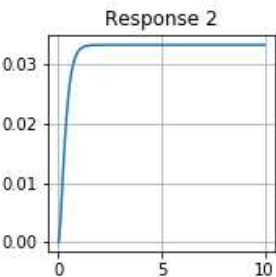
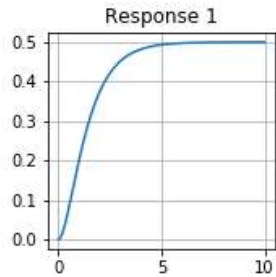


Tutorial 2: System Identification – Time Domain

Question 1

Each of the following plots 1-6 represents the impulse response for a second-order system, while the plots A-F represent their pole locations. Match each response to the corresponding pole plot.



Solutions:

Responses 1 and 2 don't appear to have any overshoot or oscillation, so we're looking for real poles. Poles A and F are options.

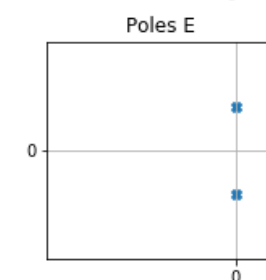
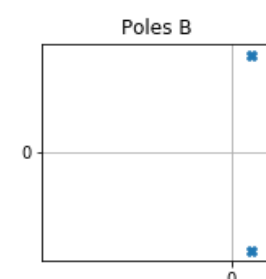
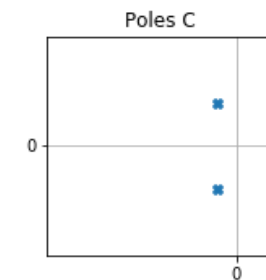
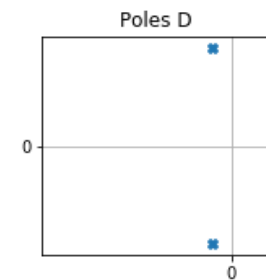
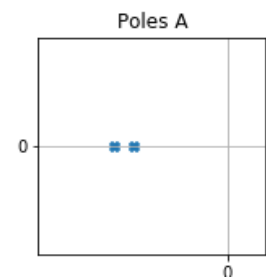
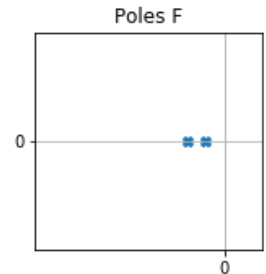
Response 1 is slower than Response 2, so we match it to F, where the poles are closer to the origin. Response 2 therefore matches A.

Responses 3 and 4 must have complex poles, because they overshoot their final values.

Pole plots C and D show stable complex pole pairs. We can match D with 3, because this response is more oscillatory, which indicates that the imaginary component of its poles would be larger. So 4 matches with C.

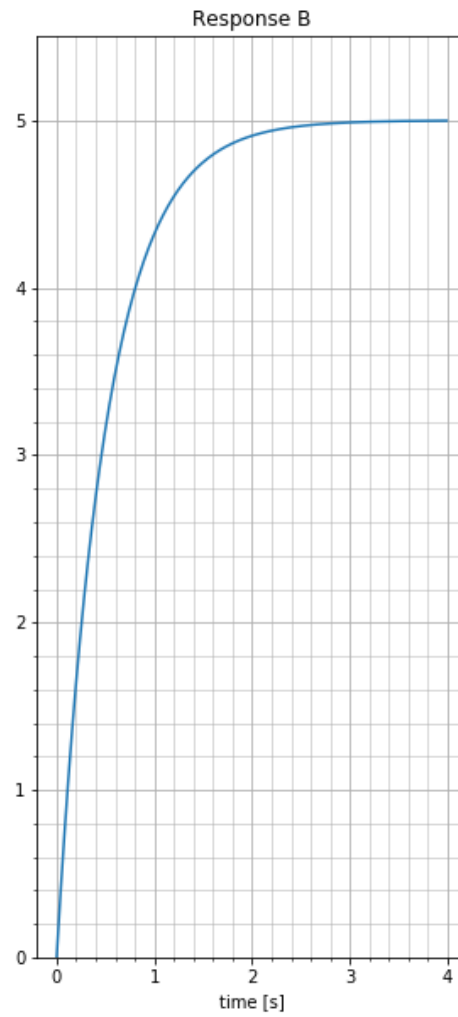
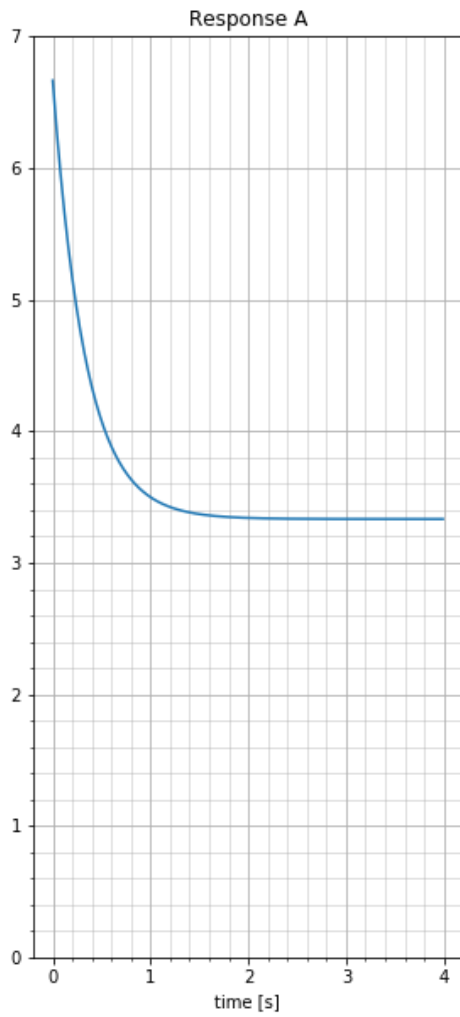
Response 5 is unstable, so it must match with B, which is the only plot showing poles in the right-half plane.

By process of elimination, this leaves E to match with Response 6. But pretending we had to think about it, the response is oscillating without growing or decaying, so we know the poles must be on the imaginary axis.



Question 2

The plots below show the step responses obtained from two unknown first-order systems. Suggest transfer functions for the systems based on these plots.



Solutions

- A) The general form of the first order transfer function we'll use for system identification is

$$\frac{A}{\tau s + 1}$$

Response A has a nonzero initial condition, so we know there must be a zero in the transfer function. So we'll add:

$$\frac{A(\frac{1}{z}s + 1)}{\tau s + 1}$$

Gain: The gridlines aren't amazing, but I eyeball that the final value is 3.35. This is $3.35 \times$ the height of the input step (1) so the gain $A = 3.35$

Time constant: To find the time constant, we need to find where the system is $1 - e^{-\frac{1}{\tau}\tau} = 1 - e^{-1} = 63\%$ of the way to its final value.

The initial value is around 6.6, so the change from initial to final value is $6.6 - 3.35 = 3.25$.

So the 63% point is $6.6 - 0.63 \times 3.25 = 4.54$, which it reaches at around 0.3 seconds.

Zero: If we fill the values we already know into the transfer function, we get

$$\frac{3.35(\frac{1}{z}s + 1)}{0.3s + 1}$$

Applying the initial value theorem to the step response gives $\lim_{s \rightarrow \infty} s \frac{1}{s} \frac{3.35(\frac{1}{z}s + 1)}{0.3s + 1} = \frac{3.35\frac{1}{z}}{0.3}$

So $\frac{3.35\frac{1}{z}}{0.3} = 6.6$ and $\frac{1}{z} = 6.6 \times \frac{0.3}{3.35} = 0.59$

Meaning the final transfer function is

$$\frac{3.35(0.59s + 1)}{0.3s + 1}$$

I accidentally saved over the code I used to generate these plots, so I can't confirm the exact original function, but I simulated the transfer function and it looks about right. (see: right)

- B) This one is a bit more straightforward because there isn't a zero.

Gain: the final value is 5, so $A = 5$

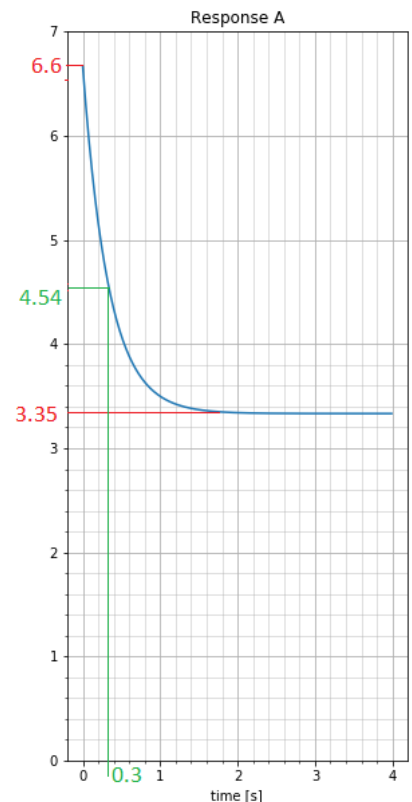
Time constant: 63% of the final value is $0.63 \times 5 = 3.15$, which it reaches at around 0.5 seconds.

So the transfer function is

$$\frac{5}{0.5s + 1}$$

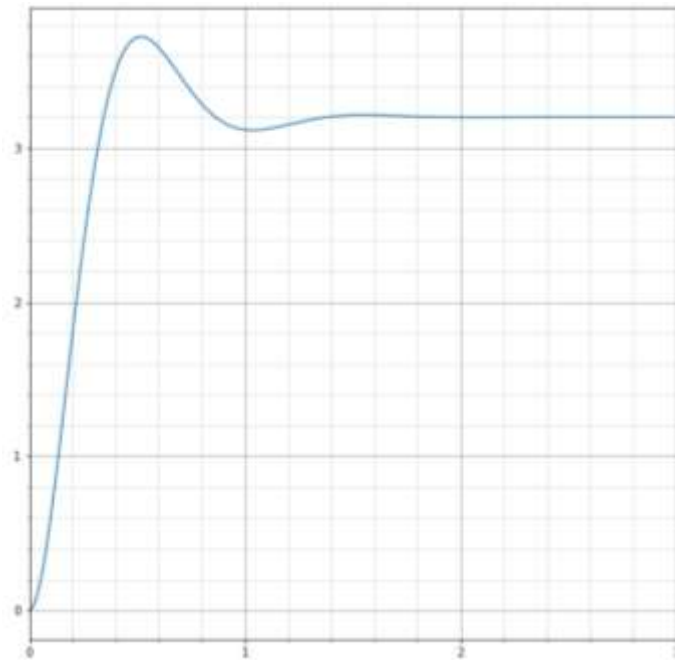
Again, simulating the result (right) indicates that

this is correct.



Question 3

- a) Find the transfer function of the second-order system that produces the following unit step response:



- b) The system is a spring mass damper with unit mass, and can be modelled by the differential equation

$$\ddot{x} = -b\dot{x} - kx + F$$

Identify the values of the spring and damping coefficients based on the transfer function.

Formulas:

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100} \sqrt{1 - \zeta^2}\right)}{-\zeta\omega_n}$$

Solutions:

- a) We are trying to find the gain A , damping ratio ζ and natural frequency ω_n to fill out the standard second order transfer function form.

The gain can be eyeballed to be $A = 3.2$ based on the final value

The only unknown in the formula for the peak value y_p is then the damping ratio ζ . From the plot, we can take $y_p = 3.7$, so

$$3.7 = 3.2 \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

Re-arrange:

$$\frac{3.7}{3.2} - 1 = 0.156 = e^{-\zeta\pi / \sqrt{1-\zeta^2}}$$

Take the natural log of both sides:

$$\log(0.156) = -1.856 = \frac{-\zeta\pi}{\sqrt{1-\zeta^2}}$$

Multiply out the denominator:

$$-1.856\sqrt{1-\zeta^2} = -\zeta\pi$$

Square both sides:

$$3.446 - 3.446\zeta^2 = \zeta^2\pi^2$$

Solve for ζ^2 :

$$\zeta^2 = \frac{3.446}{\pi^2 + 3.446} = 0.258$$

From which we find that $\zeta = 0.5$

We can use the damped frequency ω_d to find the natural frequency. Using the peak time $t_p = 0.5$, the damped frequency is $\frac{\pi}{0.5} = 6.28$

The natural frequency is then $\frac{6.28}{\sqrt{1-0.5^2}} = 7.3$

b) The characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2$ needs to match $s^2 + bs + k$

Therefore, $k = \omega_n^2 = 53.26$

and $b = 2\zeta\omega_n = 7.43$